1. Footprinting Using Search Engines and Social Networking Sites

Using the OhSINT room on TryHackMe, perform footprinting using search engines and Social networking sites.

a) Collect publicly available information about the target.
b) Categorize the collected information based on its source.

2. Website, Email, WHOIS, DNS and Network Footprinting

Using the Passive Reconnaissance room on TryHackMe, perform footprinting on an instructor-approved domain.

a) Examine domain registration and DNS records using WHOIS and DNS lookup tools.
b) Analyze the network route using traceroute and record the observations.

3. Session Hijacking Using Tools

Using the Burp Suite Basics room on TryHackMe, analyze session management in an authorized training application.

a) Inspect login and logout requests using Burp Suite.
b) Identify session cookies and examine their security attributes.

4. Detect Session Hijacking

Using Wireshark, browser Developer Tools and Burp Suite, analyze the supplied network traffic capture and simulated logs.

a) Identify suspicious session activity.
b) Differentiate normal session behavior from suspicious behavior.

5. Web Server Reconnaissance

Using the Active Reconnaissance room on TryHackMe, perform reconnaissance on an authorized lab target using Nmap and WhatWeb.

a) Identify open ports and running services.
b) Prepare a table of the detected web technologies and services.

6. Enumerate Web Server Information

Using the Network Services room on TryHackMe, enumerate the services running on an authorized lab machine.

a) Perform service detection using Nmap.
b) Compare the detected services with the expected lab configuration.

7. Web Application Reconnaissance

Using the Walking an Application room on TryHackMe, perform reconnaissance on an assigned web application.

a) Identify the pages, forms and input fields.
b) Draw an application map showing the discovered pages and navigation paths.

8. Web Spidering

Using the Content Discovery room on TryHackMe, perform web spidering on an authorized training website using OWASP ZAP or Burp Suite.

a) Examine robots.txt and sitemap.xml, where available.
b) Compare manually discovered pages with the crawler results.

9. Web Application Vulnerability Scanning

Using the OWASP Juice Shop lab on TryHackMe and OWASP ZAP, identify vulnerabilities in an authorized web application.

a) Perform a vulnerability scan and classify the findings by severity.
b) Compare automated scan results with manual browser inspection.

10. FTP Credential Dictionary Attack

Using the Network Services room on TryHackMe, perform the guided FTP dictionary attack exercise against the designated lab account.

a) Identify the FTP service and complete the guided authentication exercise.
b) Record the authentication results and analyze the effectiveness of the tested credentials.

11. Cross-Site Request Forgery (CSRF)

Using the CSRF room on TryHackMe, investigate a CSRF vulnerability in an authorized training application.

a) Identify a state-changing request and examine its CSRF protection.
b) Demonstrate the behavior of the application during the guided CSRF exercise.

12. Identify Cross-Site Scripting (XSS) Vulnerabilities

Using the XSS room on TryHackMe or the OWASP Juice Shop lab, investigate an XSS vulnerability in an authorized training application.

a) Identify input fields where user-supplied content is displayed.
b) Differentiate reflected XSS from stored XSS based on the observed application behavior.

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